

CHAPTER XXV

INDIGENOUS NARCOTICS AND STIMULANTS USED BY THE INDIANS OF THE AMAZON

[THIS chapter consists of a carefully written account of the above subject, compiled by Spruce about 1870 from his notes and observations, and printed in the short-lived *Geographical Magazine*. Fortunately, he presented the beautifully written manuscript to his Yorkshire friend and fellow-botanist, Mr. G. Stabler, of Milnthorpe, Westmoreland, who has kindly lent it me for reproduction here, and I feel sure that it will be both new and interesting to the great majority of readers of this volume. Besides its main subject, it touches upon the beliefs and customs of the Indians who use these narcotics, and on the proceedings of their "pajés" or medicine-men; and incidentally it narrates the occurrence of rare and mysterious sounds in the forest, and their very curious explanation, which I believe he was the first, and probably still the only, traveller to obtain. The whole essay affords a good example of the writer's style and of his power of making even technical details interesting, and of introducing bright descriptive flashes and touches of human nature in what might otherwise be a rather dry exposition of botanical and pharmaceutical facts. Two paragraphs

only have been omitted as unsuitable for the present work. The rest is printed verbatim, and will, I think, even to the non-botanical reader, prove not one of the least interesting chapters of this volume.]

ON SOME REMARKABLE NARCOTICS OF THE AMAZON VALLEY AND ORINOCO

In the accounts given by travellers of the festivities of the South American Indians, and of the incantations of their medicine-men, frequent mention is made of powerful drugs used to produce intoxication, or even temporary delirium. Some of these narcotics are absorbed in the form of smoke, others as snuff, and others as drink; but with the exception of tobacco, and of the fermented drinks prepared from the grain of maize, the fruit of plantains, and the roots of *Manihot utilissima*, *M. Atypi*, and a few other plants, scarcely any of them are well made out. Having had the good fortune to see the two most famous narcotics in use, and to obtain specimens of the plants that afford them sufficiently perfect to be determined botanically, I propose to record my observations on them, made on the spot.

The first of these narcotics is afforded by a climbing plant called Caapi. It belongs to the family of Malpighiaceæ, and I drew up the following brief description of it from living specimens in November 1853.

I. BANISTERIA CAAPI, Spruce

(*Pl. Exsicc.* No. 2712, *Anno* 1853)

Description.—Woody twiner; stem = thumb, swollen at joints. Leaves opposite, 6.4×3.3 , oval acuminate, apiculato-acute,

thinnish, smooth above, appresso-subpilose beneath ; on a petiole 0.9 inch long. Panicles axillary, leafy. Umbels 4-flowered. Pedicels appresso-tomentose, bracteolate only at base. Calyx deeply 5-partite ; segments ligulate, eglandulose, or with only rudimentary glands, appresso-tomentose. Petals 5, on longish thick claws ; lamina pentagonal, fimbriate, the fimbriæ clavate. Stamens 10, subunequal ; anthers roundish. Styles 3, subulate ; stigmas capitate. Capsules muricato-cristate, prolonged on one side into a greenish-white semiobovate wing (1.7 × 0.6 inch).

Habitat.—On the river Uaupés, the Içanna, and other upper tributaries of the Rio Negro, where it is commonly planted in the roças or mandioca-plots ; also at the cataracts of the Orinoco, and on its tributaries, from the Meta upwards ; and on the Napo and Pastasa and their affluents, about the eastern foot of the Equatorial Andes. Native names: Caapi, in Brazil and Venezuela ; Cadána, by the Tucáno Indians on the Uaupés ; Aya-huasca (*i.e.* Dead man's vine) in Ecuador.¹

The lower part of the stem is the part used. A quantity of this is beaten in a mortar, with water, and sometimes with the addition of a small portion of the slender roots of the Caapi-piníma.² When sufficiently triturated, it is passed through a sieve, which separates the woody fibre, and to the residue

¹ Caapí (the Portuguese have made it Caapím) is the Tupí or Lingoa Geral name for "grass." It means simply "thin leaf," and in that sense may correctly be applied to the *Banisteria Caapi*. In the same language the Maté of Paraguay (*Ilex Paraguayensis*) is called Caamirim, *i.e.* "small leaf," which is certainly not so truly said of it. The Brazilian Indians accent the last, the Venezuelan the first, syllable of Caapi.

² Caapi-piníma, *i.e.* "painted Caapi," is an Apocynous twiner of the genus *Hæmadictyon*, of which I saw only young shoots, without any flowers. The leaves are of a shining green, painted with the strong blood-red veins. It is possibly the same species as one I gathered in flower, in December 1849, at an Indian settlement on the river Trombetas (Lower Amazon), and has been distributed by Mr. Bentham under the name of *Hæmadictyon amœnicum*, n. sp. It may be the Caapi-piníma which gives its nauseous taste to the caapi drink prepared on the Uaupés, and it is probably poisonous, like most of its tribe ; but it is not essential to the narcotic effect of the *Banisteria*, which (so far as I could make out) is used without any admixture by the Guahibos, Zaporos, and other nations, out of the Uaupés.

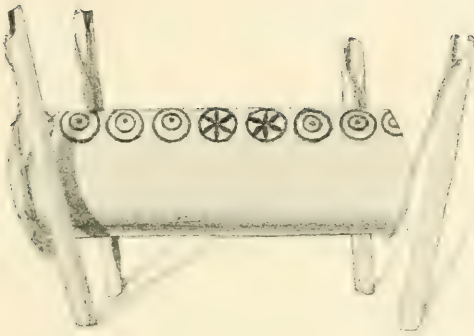
The Tucáno Indians call this plant Cadána-píra, which means the same as the Tupí name. They are the most powerful tribe on the Uaupés, and the greatest consumers of caapi ; but all the other tribes on that river—and they are about a dozen—use it in the same way.

enough water is added to render it drinkable. Thus prepared, its colour is brownish-green, and its taste bitter and disagreeable.

The Use and Effects of Caapi

In November 1852 I was present, by special invitation, at a Dabocurí or Feast of Gifts, held in a mallóca or village-house called Urubú-coará (Turkey-buzzard's nest), above the first falls of the Uaupés; the village of Panuré, where I was then residing, being at the base of the same falls, and about four miles away from Urubú-coará, following the course of the river, which during that space is a continuous succession of rapids and cataracts among rocky islands. We reached the mallóca at nightfall, just as the botútos or sacred trumpets began to boom lugubriously within the margin of the forest skirting the wide space kept open and clear of weeds around the mallóca.¹ At that sound every female outside makes a rush into the house, before the botútos emerge on the open; for to merely see one of them would be to her a sentence of death. We found about 300 people assembled, and the dances at once commenced. I need not detail the whole proceedings, for similar feasts have already been described by Mr. Wallace (*Travels on the Amazon and Rio Negro*, pp. 280 and 348). Indeed, there

¹ Some of the trumpets used at this very feast are now in the Museum of Vegetable Products at Kew. To get them out of the river Uaupés, when I left for Venezuela in March 1853, I wrapped them in mats and put them on board myself at dead of night, stowing them under the cabin floor, out of sight of my Indian mariners, who would not one of them have embarked with me had they known such articles were in the boat. The old Portuguese missionaries called these trumpets juruparís or devils—merely a bit of jealousy on their part; the botúto being the only fetish—not worshipped, but held in high respect—throughout the whole Negro-Orinoco region. (See figures opposite.)



Side view.



Back view.



End view.

FIG. 15. INDIAN SACRED DRUM OR TRUMPET.

The upright outline shows the holes at bottom of the drum (as suspended) through which the inside has been scooped out. The patterns are in red and blue.

is such a family likeness in all the Indian festivities of Tropical America that, allowing for slight local variations, the description of one might serve for all. There is no more graphic account of a native feast than that by old Wafer, of one he saw on the Isthmus of Darien (*New Voyage and Description of the Isthmus of America*, p. 363).

In the course of the night, the young men partook of caapi five or six times, in the intervals between the dances; but only a few of them at a time, and very few drank of it twice. The cup-bearer—who must be a man, for no woman can touch or taste caapi—starts at a short run from the opposite end of the house, with a small calabash containing about a teacupful of caapi in each hand, muttering “Mo-mo-mo-mo-mo” as he runs, and gradually sinking down until at last his chin nearly touches his knees, when he reaches out one of his cups to the man who stands ready to receive it, and when that is drunk off, then the other cup.

In two minutes or less after drinking it, its effects begin to be apparent. The Indian turns deadly pale, trembles in every limb, and horror is in his aspect. Suddenly contrary symptoms succeed: he bursts into a perspiration, and seems possessed with reckless fury, seizes whatever arms are at hand, his murucú, bow and arrows, or cutlass, and rushes to the doorway, where he inflicts violent blows on the ground or the doorposts, calling out all the while, “Thus would I do to mine enemy (naming him by his name) were this he!” In about ten minutes the excitement has passed off, and the Indian grows calm, but appears exhausted. Were he at home in his hut, he would sleep off the

remaining fumes, but now he must shake off his drowsiness by renewing the dance.

I had gone with the full intention of experimenting the caapi on myself, but I had scarcely dispatched one cup of the nauseous beverage, which is but half a dose, when the ruler of the feast—desirous, apparently, that I should taste all his delicacies at once—came up with a woman bearing a large calabash of caxirí (mandiocca-beer), of which I must needs take a copious draught, and as I knew the mode of its preparation, it was gulped down with secret loathing. Scarcely had I accomplished this feat when a large cigar, 2 feet long and as thick as the wrist, was put lighted into my hand, and etiquette demanded that I should take a few whiffs of it—I, who had never in my life smoked a cigar or a pipe of tobacco. Above all this, I must drink a large cup of palm-wine, and it will readily be understood that the effect of such a complex dose was a strong inclination to vomit, which was only overcome by lying down in a hammock and drinking a cup of coffee which the friend who accompanied me had taken the precaution to prepare beforehand.

White men who have partaken of caapi in the proper way concur in the account of their sensations under its influence. They feel alternations of cold and heat, fear and boldness. The sight is disturbed, and visions pass rapidly before the eyes, wherein everything gorgeous and magnificent they have heard or read of seems combined; and presently the scene changes to things uncouth and horrible. These are the general symptoms, and intelligent traders on the Upper Rio Negro, Uaupés, and

Orinoco have all told me the same tale, merely with slight personal variations. A Brazilian friend said that when he once took a full dose of caapi he saw all the marvels he had read of in the *Arabian Nights* pass rapidly before his eyes as in a panorama; but the final sensations and sights were horrible, as they always are.

At the feast of Urubú-coará I learnt that caapi was cultivated in some quantity at a roça a few hours' journey down the river, and I went there one day to get specimens of the plant, and (if possible) to purchase a sufficient quantity of the stems to be sent to England for analysis; in both which objects I was successful. There were about a dozen well-grown plants of caapi, twining up to the tree-tops along the margin of the roça, and several smaller ones. It was fortunately in flower and young fruit, and I saw, not without surprise, that it belonged to the order Malpighiaceæ and the genus *Banisteria*, of which I made it out to be an undescribed species, and therefore called it *Banisteria Caapi*. My surprise arose from the fact that there was no narcotic Malpighiad on record, nor indeed any species of that order with strong medicinal properties of any kind. *Byrsonima*—a Malpighiaceous genus that abounds in the Amazon valley—includes many species, all handsome little trees, with racemes of yellow or rose-coloured flowers, followed by small edible but rather insipid drupes. Their bark abounds in tannin, and is the usual material for tanning leather at Pará, as also, by the Indians, for dyeing coarse cotton garments a red-brown colour. Another genus—*Bunchosia*—grows chiefly on the slopes of the Andes, at from

7000 to 9000 feet elevation, and the species are trees of humble growth, bearing large yellowish-green edible drupes known as Ciruelas de fraile (Friar's plums). In cultivation the fruits are mostly seedless, and in that state are sometimes brought for sale to Ambato and other towns. The seed is described in books as poisonous, and if it be really so, then it is the only instance, so far as I know, of the existence of any hurtful principle in the entire family of Malpighiads, always excepting that of the Caapi. Yet strong poisons may lurk undiscovered in many others of the order, which is very large, and (the twining species especially) of great sameness of aspect; and the closely-allied Soapworts (Sapindaceæ) contain strong narcotic poisons, especially in the genus Paullinia.

I obtained a good many pieces of stem, dried them carefully, and packed them in a large box, which contained botanical specimens, and dispatched them down the river for England in March 1853. The man who took that box and four others on freight, in a large new boat he had built on the Uaupés, was seized for debt when about half-way down the Rio Negro, and his boat and all its contents confiscated. My boxes were thrown aside in a hut, with only the damp earth for floor, and remained there many months, when my friend Senhor Henrique Antonij, of Manáos, whom I had advised by letter of the sending-off of the boxes, heard of the mishap, and succeeded in redeeming them and getting them sent on to the port of Pará. When Mr. Bentham came to open them in England, he found the contents somewhat injured by damp and mould, and the sheets of specimens near the bottom

of the boxes quite ruined. The bundle of Caapi would presumably have quite lost its virtue from the same cause, and I do not know that it was ever analysed chemically; but some portion of it should be in the Kew Museum at this day.

Caapi is used by all the nations on the river Uaupés, some of whom speak languages differing *in toto* from each other, and have besides (in other respects) widely different customs. But on the Rio Negro, if it has ever been used, it has fallen into disuse; nor did I find it anywhere among nations of the true Carib stock, such as the Barrés, Banihuas, Mandauacas, etc., with the solitary exception of the Tarianas, who have intruded a little way within the river Uaupés, and have probably learnt to use caapi from their Tucáno neighbours.

When I was at the cataracts of the Orinoco, in June 1854, I again came upon caapi, under the same name, at an encampment of the wild Guahibos, on the savannas of Maypures. These Indians not only drink the infusion, like those of the Uaupés, but also chew the dried stem, as some people do tobacco. From them I learnt that all the native dwellers on the rivers Meta, Vichada, Guaviare, Sipapo, and the intervening smaller rivers, possess caapi, and use it in precisely the same way.

In May 1857, after a sojourn of two years in the North-Eastern Peruvian Andes, I reached, by way of the river Pastasa, the great forest of Canelos, at the foot of the volcanoes Cotopaxi, Llanganati, and Tunguragua; and in the villages of Canelos and Puca-yacu—inhabited chiefly by tribes of Zaparos—I again saw Caapi planted. It was the identical species of the Uaupés, but under a different name,

in the language of the Incas, Aya-huasca, *i.e.* Dead man's vine. The people were nearly all away at the gold-washings, but from the Governor of Puca-yacu I got an account of its properties coinciding wonderfully with what I had previously learnt in Brazil. Dr. Manuel Villavicencio, a native of Quito, who had been some years governor of the Christian settlements on the Napo, published the following year, in his *Geografia de la Republica del Ecuador* (New York, 1858), an interesting account of the customs of the natives of that river, and amongst others of their drinking the aya-huasca; but of the plant itself he could tell no more than that it was a liana or vine. The following is a summary of what I learnt at Puca-yacu and from Villavicencio of the uses and effects of the aya-huasca or caapi, as observed on the Napo and Bombonasa.

Aya-huasca is used by the Zapparos, Angutéros, Mazánes, and other tribes precisely as I saw caapi used on the Uaupés, viz. as a narcotic stimulant at their feasts. It is also drunk by the medicine-man, when called on to adjudicate in a dispute or quarrel—to give the proper answer to an embassy—to discover the plans of an enemy—to tell if strangers are coming—to ascertain if wives are unfaithful—in the case of a sick man to tell who has bewitched him, etc.

All who have partaken of it feel first vertigo; then as if they rose up into the air and were floating about. The Indians say they see beautiful lakes, woods laden with fruit, birds of brilliant plumage, etc. Soon the scene changes; they see savage beasts preparing to seize them, they can no longer hold themselves up, but fall to the ground.

At this crisis the Indian wakes up from his trance, and if he were not held down in his hammock by force, he would spring to his feet, seize his arms, and attack the first person who stood in his way. Then he becomes drowsy, and finally sleeps. If he be a medicine-man who has taken it, when he has slept off the fumes he recalls all he saw in his trance, and thereupon deduces the prophecy, divination, or what not required of him. Boys are not allowed to taste aya-huasca before they reach puberty, nor women at any age: precisely as on the Uaupés.

Villavicencio says (*op. cit.* p. 373): "When I have partaken of aya-huasca, my head has immediately begun to swim, then I have seemed to enter on an aerial voyage, wherein I thought I saw the most charming landscapes, great cities, lofty towers, beautiful parks, and other delightful things. Then all at once I found myself deserted in a forest and attacked by beasts of prey, against which I tried to defend myself. Lastly, I began to come round, but with a feeling of excessive drowsiness, headache, and sometimes general *malaise*."

This is all I have seen and learnt of caapi or aya-huasca. I regret being unable to tell what is the peculiar narcotic principle that produces such extraordinary effects. Opium and hemp are its most obvious analogues, but caapi would seem to operate on the nervous system far more rapidly and violently than either. Some traveller who may follow my steps, with greater resources at his command, will, it is to be hoped, be able to bring away materials adequate for the complete analysis of this curious plant.

Niopo Snuff and the Mode of using it

II. PIPTADENIA NIOPO, Humboldt

Synonyms—*Acacia*? *Niopo*, Humb., *Rel. Hist.* ii. p. 620: ejusdem *Nov. Gen. Amer.* vi. p. 282; *DC. Prodr.* ii. p. 471. *Inga Niopo*, Willd.

Description.—Tree, 50 feet by 2 feet, with muricated bark; otherwise unarmed. Leaves bipinnate; pinnæ twenty-four pairs; pinnules very numerous, minute, linear, mucronato-apiculate, ciliated, sparsely sub-pubescent. An oblong gland on petiole above base; another between terminal pinnæ. Racemes axillary and terminal; pedicels twin, each bearing a small globose head of white flowers. Corolla slightly emerged from 5-angled calyx. Stamens 10; anthers tipped with a gland. Pod linear, sub-compressed, apiculate, 7-12-seeded, sub-constricted between seeds. Seeds flattish, green.

Habitat.—In the drier forests of the Amazon, and along its tributaries, both northern and southern; on the Rio Negro, throughout its course; also at the cataracts of the Orinoco; both wild and planted near villages. (Santarem, fl. Amazonum, Spruce, *Exsicc.* No. 828, etiam Janauari, fl. Negro, No. 1786.) Native names: Paricá in Brazil; Niopo in Venezuela.

We owe our first knowledge of Niopo snuff, and of the tree producing it, to Humboldt and Bonpland, whose brief account of it is thus condensed by Kunth: "Ex seminibus tritis calci vivae admixtis fit tabacum nobile quo Indi Otomacos et Guajibos utuntur" (*Synopsis*, iv. p. 20). In the modern niopo, as I saw it prepared by the Guahibos themselves, there is no admixture of quicklime, and that is the sole difference. My specimens of the leaves, flowers, and fruit agree so well with Kunth's description of *Acacia Niopo* that I cannot doubt their being the same species; especially as I have traced the tree all the way from the Amazon to the Orinoco, and found it everywhere identical, although it bears a different name on the two rivers, as is commonly the case where the same plant or animal occurs on

both. Mr. Bentham believes my plant to be the old *Mimosa peregrina* of Linnæus (*Acacia peregrina*, Willd.); and if both opinions be correct, then the species must be called *Piptadenia peregrina* (L.), Benth.; and *Acacia Niopo*, Humb., will stand as a synonym.

I first gathered specimens of the Parica (or Niopo) tree in 1850, near Santarem, at the junction of the Tapajoz and Amazon, where it had apparently been planted. In the following year I gathered it on the little river Jauauarí—one of the lower tributaries of the Rio Negro—where it was certainly wild. But I did not see the snuff actually prepared from the seeds and in use until June 1854, at the cataracts of the Orinoco. A wandering horde of Guahibo Indians, from the river Meta, was encamped on the savannas of Maypures, and on a visit to their camp I saw an old man grinding Niopo seeds, and purchased of him his apparatus for making and taking the snuff, which is now in the Museum of Vegetable Products at Kew. I proceed to describe both processes.

The seeds being first roasted, are powdered on a wooden platter, nearly the shape of a watch-glass, but rather longer than broad ($9\frac{1}{4}$ inches by 8 inches). It is held on the knee by a broad thin handle, which is grasped in the left hand, while the fingers of the right hold a small spatula or pestle of the hard wood of the Palo de arco (*Tecomæ* sp.) with which the seeds are crushed.

The snuff is kept in a mull made of a bit of the leg-bone of the jaguar, closed at one end with pitch, and at the other end stopped with a cork of marima bark. It hangs around the neck, and from it are

suspended a few odoriferous rhizomes of a sedge (*Kyllingia odorata*). Rhizomes of the same sedge, or of an allied species, are in use among the Indians throughout the Amazon and Orinoco. They render the wearer secure from the bad wish and evil eye of his enemies.

For taking the snuff they use an apparatus made of the leg-bones of herons or other long-shanked birds put together in the shape of the letter Y, or something like a tuning-fork, and the two upper tubes are tipped with small black perforated knobs (the endocarps of a palm). The lower tube being inserted in the snuff-box and the knobs in the nostrils, the snuff is forcibly inhaled, with the effect of thoroughly narcotising a novice, or indeed a practised hand, if taken in sufficient quantity; but this endures only a few minutes, and is followed by a soothing influence, which is more lasting.

The Guahibo had a bit of caapi hung from his neck, along with the snuff-box, and as he ground his niopo he every now and then tore off a strip of caapi with his teeth and chewed it with evident satisfaction. "With a chew of caapi and a pinch of niopo," said he, in his broken Spanish, "one feels so good! No hunger—no thirst—no tired!" From the same man I learnt that caapi and niopo were used by all the nations on the upper tributaries of the Orinoco, *i.e.* on the Guaviare, Vichada, Meta, Sipapo, etc.

I had previously (in 1851) purchased of a Brazilian trader at Manáos an apparatus for taking niopo snuff rather different from that of the Guahibos. He had brought it from the river Purús, where it had been used by the Catauixí Indians. My note

on it (as taken down from his account) is as follows:—

The Catauixís use niopo snuff as a narcotic stimulant, precisely as the Guahibos of Venezuela, and as the Muras and other nations of the Amazon, where it is called paricá. For absorbing paricá by the nose, a bent tube is made of a bird's shank-bone, cut in two, and the pieces joined by wrapping, at such an angle that one end being applied to the mouth, the other reaches the nostrils. A portion of snuff is then put into the tube and blown with great force up the nose. A clyster-pipe is made, on the same principle, of the long shank-bone of the tuyuyú (*Mycteria americana*). The effect of paricá, taken as snuff, is to speedily induce a sort of intoxication, resembling in its symptoms (as described to me in this instance) that produced by the fungus *Amanita muscaria*. Taken in injection, it is a purge, more or less violent according to the dose. When the Catauixí is about to set forth on the chase, he takes a small injection of paricá, and administers another to his dog, the effect on both being (it is said) to clear their vision and render them more alert!

Herndon (*Valley of the Amazon*, p. 318) gives the following account of the use of paricá among the Mundrucús, on the river Tapajoz, which he derived from an intelligent Frenchman (M. Maugin) who had traded among them. They powder the seeds of paricá, make the powder into a paste, and repulverise a portion whenever they want to take it as snuff. Two quills of the royal heron, joined side by side, make a double tube, which is applied to the nostrils and the powder snuffed up with

a strong inspiration. M. Maugin thus describes its effects on an Indian whom he saw take it. "His eyes started from his head, his mouth contracted, his limbs trembled. It was fearful to see him. He was obliged to sit down or he would have fallen. He was drunk, but only for about five minutes; he was then gayer."

"Medicine-Men" and their Customs

Among the native tribes of the Uaupés and of the upper tributaries of the Orinoco, niopo or paricá is the chief curative agent. When the payé is called in to treat a patient, he first snuffs up his nose such a quantity of paricá as suffices to throw him into a sort of ecstasy, wherein he professes to divine the nature of the evil wish which has caused the sickness, and to gather force to counteract it. He next lights a very thick cigar of tobacco, inhales a quantity of smoke, and puffs it out over the sick man, over the hammock in which he is laid, and over everything he habitually uses, but especially over the food he is to eat. This done, the payé professes to suck out the ill, by applying his mouth to the seat of pain, or as near to it as practicable; and he spits out the morbid matter—most likely tobacco or coca juice—and sometimes produces from his mouth thorns and other substances, previously hidden there, but which he pretends to have extracted from the sick man's body. If the sickness ends fatally, he denounces the enemy whose evil wish has caused it, and not infrequently it is some rival payé, of the same or another nation. Hence I was told that the payés

never travel without an accompaniment of at least four or five well-armed men, their lives being in continual jeopardy from such denunciations.

I have never been so fortunate as to see a genuine payé at work. Among the civilised Indians the Christian padre has supplanted the pagan payé, who has besides been discountenanced and persecuted by the civil authorities; so that if any now exist, he must exercise his office in secret. With the native and still unchristianised tribes I have for the most part held only passing intercourse during some of my voyages. Once I lived for seven months at a time among them, on the river Uaupés, but even there I failed to catch a payé. When I was exploring the Jauarité cataracts on that river, and was the guest of Uiáca, the venerable chief of the Tucáno nation, news came to the mallóca one afternoon that a famous payé, from a long way up river, would arrive that night and remain until next day, and I congratulated myself on so fine a chance of getting to know some of the secrets of his "medicine." He did not reach the port until 10 P.M., and when he learnt that there was a white payé (meaning myself) in the village, he and his attendants immediately threw back into the canoe his goods, which they had begun to disembark, and resumed their dangerous voyage down the river in the night-time. I was told he had with him several palm-leaf boxes, containing his apparatus. (There is a similar box now in the Kew Museum, sent by me from the Uaupés.) I could only regret that his dread of a supposed rival had prevented the interview which to me would have been full of interest; the more so as I was prepared to barter

with him for the whole of his materia medica, if my stock-in-trade would have sufficed.

Rocheport (*Histoire Naturelle et Morale des Isles Antilles*, Rotterdam, 1665), says: "Their Boyés or medicine-men practise both medicine and devilry. They are resorted to: 1, to cause punishment to fall on some one who has wronged or injured the applicant; 2, to cure some disease; 3, to foretell the advent of a war; 4, to drive out the Maboya or Evil Spirit" (p. 472).

Their functions are very much the same at the present day among the native tribes of the mainland as they were two or three hundred years ago in the isles of the Caribbean Sea. I propose, in what follows, to review briefly the use made by the payés of their materia medica in the treatment of disease.

The apparatus and materia medica of the medicine-men of the region lying adjacent to the Upper Rio Negro and Orinoco, and extending thence westward to the Andes, are chiefly the following:—

The Maraca or Rattle.

Tobacco, juice and smoke.

Niopo (or Paricá), powdered seeds in snuff.

Caapi (or Aya-huasca), stems in infusion.

1. *The Maraca or Rattle*.—This is the hard globose or oval pericarp of the *Crescentia Cujete*, or sometimes of a gourd, tastefully engraved and perforated in geometrical or fantastic designs, and the lines usually coloured. To make it rattle, a few small bright-red or red-and-black beans are put into it; those most used on the Uaupés are seeds of *Batesia erythrosperma* (Spruce) and of *Ormosia coccinea* (Jack). I have seen the maraca used in

dances, but it is also employed by the payés in their divinations, and Bancroft's account of its use in Guayana corresponds so nearly with what was told to me on the Uaupés, that I cannot do better than transcribe it here.

“The medicine-men, called Peiís [Stedman says Peiis or Pagayers], unite in themselves the sacerdotal and medicinal functions. One of the implements of the peii is a hollowed calabash (cuya) through the centre of which an axis is passed projecting about a foot on each side, the thick end forming a handle, the thin end decorated with feathers; it is also carved and painted and perforated with small holes—some long, some round—and several quartz pebbles and red-and-black beans are put inside it, so that it forms a rattle. When the peii is called to a patient, he begins his exorcism at night, the lights being put out and he left alone with the patient. He rattles his maraca by turning it slowly round, singing at the same time a supplication to the Yawahoo. This goes on for say a couple of hours, when the peii is heard conversing with the Yawahoo—at least there are two distinct voices. Afterwards the peii makes a report in an ambiguous style, on what will be the event of the disorder. The exorcisms are repeated every night until after a favourable turn, when the peii pretends to extract the cause of the disorder by sucking the part affected, after which he pulls out of his mouth fish-bones, thorns, snake's teeth, or some such substance, which he has before concealed therein, but pretends to have been maliciously conveyed into the affected part by the Yawahoo. The patient then fancies himself cured, and the influence